SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Claypot Chicken Rice
Project / Solution Name	Jobee
Name (Leader)	Lam Way Hou
Email & Phone (Leader)	lamwayhou@gmail.com +60162211236
University Affiliation(s)	Sunway University, Universiti Putra Malaysia
Live Prototype URL	https://jobee-sdchack.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/jimmerino2/sdc_hackathon_2026

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Learners use a Vue 3 single-page app, served with our FastAPI backend from one Replit Autoscale deployment. The app calls the backend over REST, and long tasks stream live progress. Supabase Postgres stores profiles, roadmaps and reference data, and Supabase Auth handles emailed one-time-code login. The backend sends prompts through an OpenAI-compatible gateway to a pre-trained language model, which suggests resume skills, plans roadmaps, writes and judges starter projects, and explains role matches. A documented formula, never the model, ranks roles.

For AI-assisted learning, the model writes one AI skill per roadmap section, teaching learners to use AI tools to learn that section's skills. It comes first and unlocks the rest. AI skills are stored by role for reuse, and source links come from ESCO, the European Commission's skills taxonomy, never the model. Roadmaps still build if the model or ESCO fails.

Offline Python scripts clean and join public datasets (Indeed Hiring Lab, O*NET and others) into forecast tables the backend reads. On the Signal page, our LightGBM model retrains in memory on learner-selected quarters and stores the result in Supabase. Optional connections: the public HRD Corp programme list and one-time LinkedIn OAuth sharing that keeps no token.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12+ (3.14.1 used in development)	Backend language	PSF License
TypeScript 5.9.3	Frontend language (typed JavaScript)	Apache-2.0
Vue 3.5.44	Frontend framework for the single-page app	MIT

Name & Version	Purpose in Project	License
Vue Router 5.4.0	Page routing and the onboarding flow guard	MIT
Vite 7.3.7	Frontend build tool and dev server	MIT
@vitejs/plugin-vue 6.0.10	Vue support inside Vite	MIT
vue-tsc 3.3.13	Type-checking for Vue files	MIT
Vitest 4.1.12	Frontend tests	MIT
lucide-vue-next 1.0.0	Icons	ISC
pdfjs-dist 3.11.174 (PDF.js)	Reading uploaded resume PDFs in the browser	Apache-2.0
FastAPI 0.153.0	Backend web framework (REST and streaming)	MIT
Uvicorn 0.54.0	Server that runs FastAPI	BSD-3-Clause
Pydantic 2.14.1	Request and response validation	MIT
pydantic-settings 2.16.1	Reads configuration from environment	MIT
supabase-py 2.33.0	Database and login client	MIT
openai (Python SDK) 3.35.1	Calls the LLM gateway (OpenAI-compatible)	Apache-2.0
LightGBM 4.7.0	Gradient-boosted forecasting model	MIT
scikit-learn 1.9.1	Forecast evaluation and preprocessing	BSD-3-Clause
pandas 3.0.6	Data tables and the forecast pipeline	BSD-3-Clause

Name & Version	Purpose in Project	License
NumPy 2.5.4	Numerical computation	BSD-3-Clause
pypdf 6.21.1	Reading text from uploaded PDFs on the server	BSD-3-Clause
openpyxl 3.1.5	Reading Excel data files in the pipeline	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Supabase (Postgres and Auth)	Database for profiles, roadmaps and reference data; passwordless email login codes	free
LLM gateway, litellm.rapidscreen.io (KrackedDevs), serving DeepSeek V4.1 Flash	Resume skill extraction, roadmap planning, starter projects and judging, role-interest reasons	sponsor-provided
GitHub REST API (GitHub)	Reading a learner's submitted repository to judge a challenge	free
LinkedIn API: Sign In with OpenID Connect and Share (LinkedIn)	Optional one-time sign-in and post to share a completed roadmap	free
Google Fonts (Google)	Nunito typeface in the app and emails	free
Indeed Hiring Lab job_postings_tracker and ai-tracker (GitHub data repositories)	Job-posting volumes and AI share of postings for the forecasts	free
O*NET Database 31.0 (U.S. Department of Labor)	Occupations, skills and technology lists	free
Hacker News Algolia Search API (Algolia / Y Combinator)	Monthly 'Who is hiring?' ads for skill-demand series and emerging skills	free
Adzuna Jobs API (Adzuna)	Job-ad snapshots as a cross-check only, not used in models	free
FRED / ALFRED API (Federal Reserve Bank of St. Louis)	Real-Time Population Survey series on worker generative-AI use	free
U.S. Census Bureau BTOS data downloads	Business AI-adoption rate	free

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Epoch AI benchmark and models data (epoch.ai) and METR benchmark results (metr.org)	AI capability and time-horizon data for the AI outlook	free
Wayback Machine CDX API (Internet Archive)	Older dated copies of Epoch's data for backtesting	free
Anthropic Economic Index (Hugging Face)	Task exposure to AI	free
DOSM Job Market Insights (Department of Statistics Malaysia)	Malaysian vacancies by occupation group	free
HRD Corp BizMatch public programme list (HRD Corp)	Claimable training programmes shown beside skills (collected by script and linked back)	free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Hosts the live prototype: serves the Vue frontend and the FastAPI backend in one deployment, and scales with traffic	Region not mentioned
Supabase	Postgres database for profiles, roadmaps, reference data, forecast signals and HRD Corp programmes	Free Plan, ap-southeast-1 (Singapore)
Supabase Auth	Passwordless email one-time-code login	Free Plan, ap-southeast-1 (Singapore)
KrackedDevs	Routes all model calls (resume skills, roadmaps, projects, judging) to the language model; compute for the AI features	Region not mentioned

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
DeepSeek V4.1 Flash (model id deepseek-v4.1-flash)	DeepSeek, called through the KrackedDevs LiteLLM gateway (litellm.rapidscreen.io)	Pre-trained, used as is (no fine-tuning); prompted for resume skill extraction, roadmap planning, project writing and judging, and role-interest reasons	MIT
LightGBM gradient-boosted trees, forecasting model (LightGBM library 4.7.0)	Self-built using the open-source LightGBM library	Self-trained on Indeed Hiring Lab postings and AI-tracker data; retrains in memory when the user selects quarters; shown as a comparison line, since simple baselines beat it in backtesting	Our own model; LightGBM library is MIT

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Indeed Job Postings Index	https://github.com/hiring-lab/job_postings_tracker	Public	CC-BY-4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Indeed AI Tracker	https://github.com/hiring-lab/ai-tracker	Public	CC-BY-4.0	N
O*NET 31.0 Database (roles, skills, technology)	https://www.onetcenter.org/database.html	Public	CC-BY-4.0	N
Anthropic Economic Index	https://huggingface.co/datasets/Anthropic/EconomicIndex	Public	CC-BY (version to confirm)	N
DOSM Job Market Insights, Malaysia (Q2 2026)	https://www.dosm.gov.my/portal-main/release-content/big-data-analytics-job-market-insights-q22026	Public	Not stated on release page	N
Census Business Trends and Outlook Survey (BTOS)	https://www.census.gov/hfp/btos/data_downloads	Public	Public domain (US government work)	N
Real-Time Population Survey on generative AI use (via FRED)	https://sites.google.com/view/covid-rps/data	Public	Not stated, to confirm	N
Epoch AI benchmark data and AI models data	https://epoch.ai/benchmarks and https://epoch.ai/data/ai-models	Public	CC-BY-4.0	N
METR time-horizon benchmark results	https://metr.org/assets/benchmark_results_1_1.yaml	Public	Not stated, to confirm	N
Hacker News "Ask HN: Who is hiring?" threads	https://hn.algolia.com/api	Public	Public API terms	N (company job ads)

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Stack Overflow Developer Survey microdata 2023-2025	https://survey.stackoverflow.co	Public	ODbL	N (anonymised responses)
Adzuna job-ads snapshots (context only, not used in models)	Adzuna API (free tier)	Public	Adzuna API terms	N
HRD Corp BizMatch programme list	https://bizmatch.hrdcorp.gov.my	Public	All rights reserved (linked to, not redistributed)	N (training providers are companies)
Learning resource links per skill (up to 3 per skill)	Collected by web search on 22 Oct 2026; links to YouTube, courses and similar sites	Self-collected	Links only; content stays with each site	N
Job-to-Indeed-sector crosswalk, Malaysia occupation-group map, SDG weights and targets for roles and skills, role goals, interests	Built by our team (pipeline/ and data/processed/ in the repository)	Self-collected	Our own work	N
Learner profiles (name, email, stage, interests, skills, roadmap progress)	Created by users in the running app and stored in Supabase	Private	Not shared; see Privacy Policy	Y (name and email)
Uploaded resumes	Uploaded by users; text read in memory, sent to the AI model, not stored	Private	Not shared; see Privacy Policy	Y (may contain personal details)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Sonnet 5.5 Claude Opus 5.5 Claude Code Antigravity	Supportive coding tool to turn our prompts and logics into detailed codes and mathematical equations Generating UI prototypes and final designs based on revisions and initial layout wireframes we created	Moderate/Major
Replit	Setting up and publishing the Replit development, running the app, and cleanup changes on the deployment branches	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every team member reviewed AI-generated code before committing it, and each commit was made under a named team member. We ran automated checks on the changes: backend tests, frontend tests, TypeScript type-checking, and a production build. We then ran the app and checked the screens ourselves, including the data refresh and role matching. We kept AI out of the role ranking, which uses a documented formula. AI-written text, such as the privacy policy and cost estimates, was checked against the code. We also kept the data sources and licences ourselves. The team is accountable for all submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Hosting / platform (sponsor-provided)	Hosts the live prototype (jobee-sdchack.replit.app), serving the Vue frontend and the FastAPI backend, with Replit Agent used during development
RunPod	GPU/compute credits/Serverless	Hosted the self-run resume scoring model (OpenJev). We have since removed OpenJev from the app and shut the endpoint down due to credits are used up
KrackedDevs	LiteLLM gateway access (litellm.rapidscreen.io)	Serves the language model for all AI features in the current build [confirm it was sponsor-provided]

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replaceable. The AI gateway powers resume reading, roadmaps and projects, but the app talks to it through a standard OpenAI-compatible interface, so any other provider works by changing a URL and key. Role ranking, forecasts and the dashboards need no AI model, so they run without it. Replit hosting is also replaceable: the app runs locally and has a Vercel configuration. RunPod is no longer used; we removed it when the credits ran out, with no loss of core functionality.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Jobee stores a learner's chosen name, email address, career stage, interests, skills and roadmap progress in Supabase. Login uses an emailed one-time code, so we hold no passwords. Uploaded resumes are read in memory and never saved; their text is sent to the AI gateway to suggest skills, along with profile details when roadmaps are built. LinkedIn sharing is optional, and its permission is used once and not stored. Data travels over HTTPS and sits in a managed database with access rules. Learners must tick a consent box before giving their email, and a Privacy Policy explains all of this. Data is kept until the learner asks for removal, which our team handles by hand. The AI provider's own retention is outside our control.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The language model can invent skills, links or advice. We limit this by matching resume skills to a fixed catalogue, letting learners edit the suggested skills, showing sources and dates, and keeping the AI out of role ranking, which uses a published formula. Forecasts are uncertain: LightGBM did not beat simple baselines in backtests, so the headline shows last year's pace. The three AI futures are labelled assumptions, and US data stands in for Malaysia. Time estimates are mostly formula-based. AI skills are model-written and not human-reviewed, and reuse can spread a poor one across a role's learners. Links come only from ESCO. Learners may over-rely on AI. Bias has not been formally tested; a fairness check is planned. We depend on one AI gateway with rate limits; the code uses a standard OpenAI-compatible interface, so the provider can be swapped by changing a URL and key.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



01/11/2026